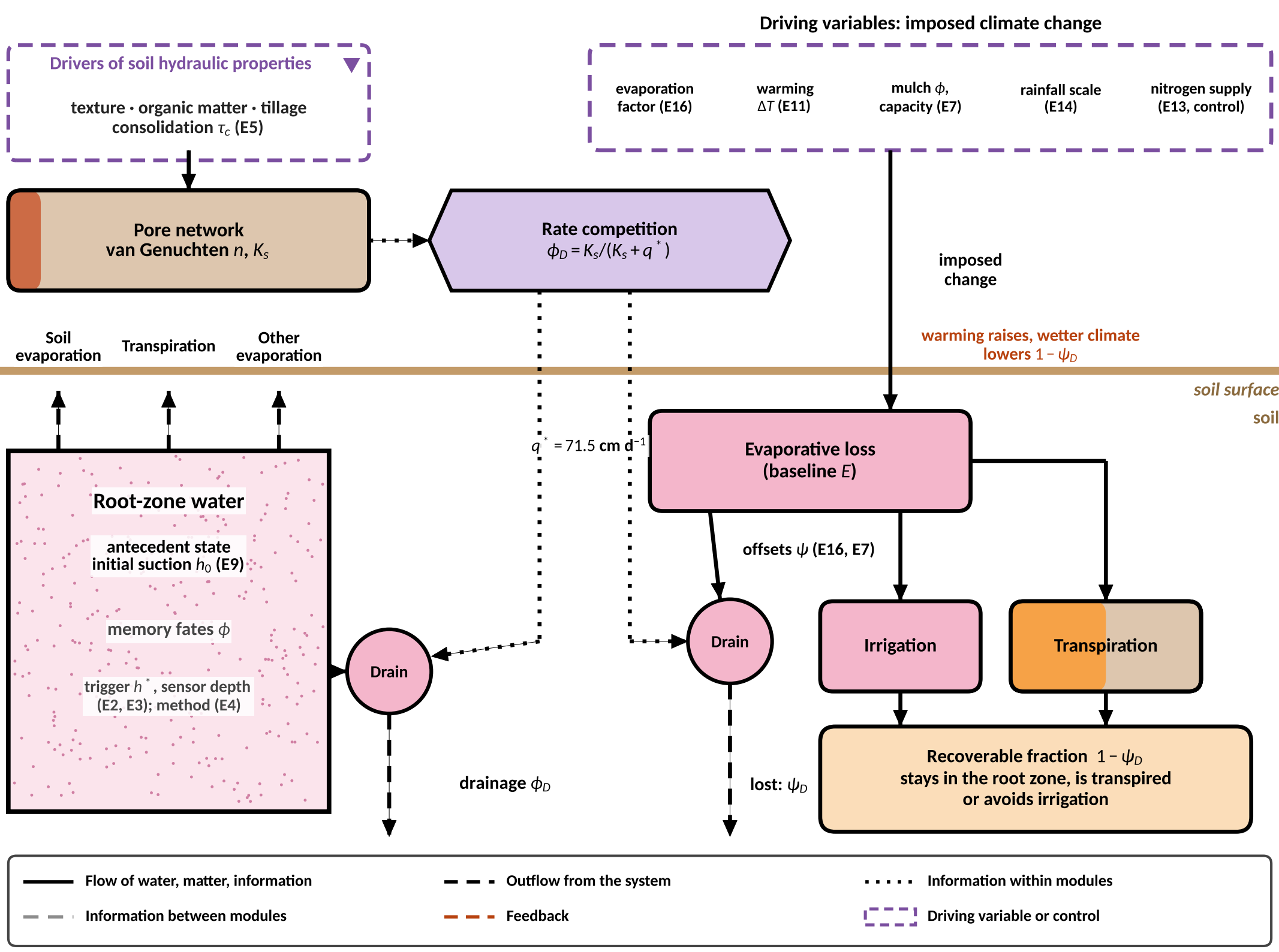

Driving variables: imposed climate change
Drivers of soil hydraulic properties
texture · organic matter · tillage
consolidation τ_c (E5)
evaporation factor (E16)
warming ΔT (E11)
mulch ϕ, capacity (E7)
rainfall scale (E14)
nitrogen supply (E13, control)
Pore network
van Genuchten n, K_s
Rate competition
$\phi_D = K_s/(K_s + q^*)$
imposed change
Soil evaporation
Transpiration
Other evaporation
warming raises, wetter climate lowers $1 - \psi_D$
soil surface
soil
$q^* = 71.5$ cm d^{-1}
Evaporative loss (baseline E)
Root-zone water
antecedent state
initial suction h_0 (E9)
memory fates ϕ
trigger h^*, sensor depth (E2, E3); method (E4)
offsets ψ (E16, E7)
Drain
Drain
Irrigation
Transpiration
drainage ϕ_D
lost: ψ_D
Recoverable fraction $1 - \psi_D$
stays in the root zone, is transpired or avoids irrigation
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Driving variable or control